

Data Cleaning Project

```
In [1]: import numpy as np  
import pandas as pd
```

```
In [2]: pd.__version__
```

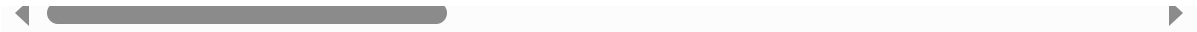
```
Out[2]: '2.2.3'
```

```
In [3]: df = pd.read_csv('E:/Data Analytics Python Projects/Pandas Project/Pandas/Airbnb_Op
```

```
In [4]: df
```

Out[4]:

	id	NAME	host id	host_identity_verified	host name	neighborhood
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan
3	1002755	NaN	85098326012	unconfirmed	Garry	Brooklyn
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan
...	...	...	...	...	...	...
102594	6092437	Spare room in Williamsburg	12312296767	verified	Krik	Brooklyn
102595	6092990	Best Location near Columbia U	77864383453	unconfirmed	Mifan	Manhattan
102596	6093542	Comfy, bright room in Brooklyn	69050334417	unconfirmed	Megan	Brooklyn
102597	6094094	Big Studio-One Stop from Midtown	11160591270	unconfirmed	Christopher	Manhattan
102598	6094647	585 sf Luxury Studio	68170633372	unconfirmed	Rebecca	Manhattan
102599 rows × 26 columns						



```
In [5]: df.head()
```

Out[5]:

	id	NAME	host id	host_identity_verified	host name	neighbourhood group
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan
3	1002755	NaN	85098326012	unconfirmed	Garry	Brooklyn
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan

5 rows × 26 columns



```
In [6]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 102599 entries, 0 to 102598
Data columns (total 26 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                    102599 non-null  object
1   NAME                                102349 non-null  object
2   host id                             102599 non-null  object
3   host_identity_verified               102310 non-null  object
4   host name                           102193 non-null  object
5   neighbourhood group                 102570 non-null  object
6   neighbourhood                       102583 non-null  object
7   lat                                 102591 non-null  object
8   long                                102591 non-null  object
9   country                             102067 non-null  object
10  country code                        102468 non-null  object
11  instant_bookable                    102494 non-null  object
12  cancellation_policy                 102523 non-null  object
13  room type                           102599 non-null  object
14  Construction year                   102385 non-null  object
15  price                               102352 non-null  object
16  service fee                         102326 non-null  object
17  minimum nights                      102190 non-null  object
18  number of reviews                   102416 non-null  object
19  last review                         86706 non-null   object
20  reviews per month                   86720 non-null   object
21  review rate number                  102273 non-null  object
22  calculated host listings count       102280 non-null  object
23  availability 365                    102151 non-null  object
24  house_rules                         50468 non-null   object
25  license                             2 non-null       object
dtypes: object(26)
memory usage: 20.4+ MB

```

```
In [7]: df["license"]
```

```

Out[7]: 0      NaN
        1      NaN
        2      NaN
        3      NaN
        4      NaN
        ...
102594  NaN
102595  NaN
102596  NaN
102597  NaN
102598  NaN
Name: license, Length: 102599, dtype: object

```

Deleting redundant columns

```
In [8]: df.columns
```

```
Out[8]: Index(['id', 'NAME', 'host id', 'host_identity_verified', 'host name',  
             'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',  
             'country code', 'instant_bookable', 'cancellation_policy', 'room type',  
             'Construction year', 'price', 'service fee', 'minimum nights',  
             'number of reviews', 'last review', 'reviews per month',  
             'review rate number', 'calculated host listings count',  
             'availability 365', 'house_rules', 'license'],  
            dtype='object')
```

```
In [9]: columns_to_keep = ['NAME', 'host id', 'host_identity_verified', 'host name',  
                          'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',  
                          'country code', 'instant_bookable', 'cancellation_policy', 'room type',  
                          'Construction year', 'price', 'service fee', 'minimum nights',  
                          'number of reviews', 'last review']  
  
columns_to_drop = ['id', 'reviews per month',  
                  'review rate number', 'calculated host listings count',  
                  'availability 365', 'house_rules', 'license']
```

```
In [10]: len(columns_to_keep)
```

```
Out[10]: 19
```

```
In [11]: len(columns_to_drop)
```

```
Out[11]: 7
```

Filtering Data

```
In [12]: df[columns_to_keep]
```

Out[12]:

	NAME	host id	host_identity_verified	host name	neighbourhood group	n
0	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	
1	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	
2	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	
3	NaN	85098326012	unconfirmed	Garry	Brooklyn	
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	
...	...	...	...	...	...	...
102594	Spare room in Williamsburg	12312296767	verified	Krik	Brooklyn	
102595	Best Location near Columbia U	77864383453	unconfirmed	Mifan	Manhattan	
102596	Comfy, bright room in Brooklyn	69050334417	unconfirmed	Megan	Brooklyn	
102597	Big Studio-One Stop from Midtown	11160591270	unconfirmed	Christopher	Queens	L
102598	585 sf Luxury Studio	68170633372	unconfirmed	Rebecca	Manhattan	

102599 rows × 19 columns



```
In [13]: df.shape
```

Out[13]: (102599, 26)

```
In [14]: df1 = df.drop(columns= columns_to_drop)
```

```
In [15]: df1
```

Out[15]:

	NAME	host id	host_identity_verified	host name	neighbourhood group	n
0	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	
1	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	
2	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	
3	NaN	85098326012	unconfirmed	Garry	Brooklyn	
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	
...	...	...	...	...	...	...
102594	Spare room in Williamsburg	12312296767	verified	Krik	Brooklyn	
102595	Best Location near Columbia U	77864383453	unconfirmed	Mifan	Manhattan	
102596	Comfy, bright room in Brooklyn	69050334417	unconfirmed	Megan	Brooklyn	
102597	Big Studio-One Stop from Midtown	11160591270	unconfirmed	Christopher	Queens	L
102598	585 sf Luxury Studio	68170633372	unconfirmed	Rebecca	Manhattan	

102599 rows × 19 columns



```
In [16]: df1.shape
```

Out[16]: (102599, 19)

```
In [17]: df1.columns
```

```
Out[17]: Index(['NAME', 'host id', 'host_identity_verified', 'host name',
              'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
              'country code', 'instant_bookable', 'cancellation_policy', 'room type',
              'Construction year', 'price', 'service fee', 'minimum nights',
              'number of reviews', 'last review'],
              dtype='object')
```

Renaming the columns

```
In [18]: df1.rename(columns = {'NAME' : 'Name'} , inplace=True)
```

```
In [19]: df1.head()
```

Out[19]:

	Name	host id	host_identity_verified	host name	neighbourhood group	neighbourhood
0	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington
1	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown
2	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem
3	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem

```
In [20]: # Converting column names in lower case

new_column_name = []
for i in df1.columns:
    new_column_name.append(i.lower())
```

```
In [21]: new_column_name
```



```
Out[21]: ['name',
          'host id',
          'host_identity_verified',
          'host name',
          'neighbourhood group',
          'neighbourhood',
          'lat',
          'long',
          'country',
          'country code',
          'instant_bookable',
          'cancellation_policy',
          'room type',
          'construction year',
          'price',
          'service fee',
          'minimum nights',
          'number of reviews',
          'last review']
```

```
In [22]: # Naming columns with upper case
df1.columns = new_column_name
```

```
In [23]: df1.head()
```

Out[23]:

	name	host id	host_identity_verified	host name	neighbourhood group	neighbourhood
0	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington
1	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown
2	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	Harlem
3	NaN	85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem

Checking Duplicate data

```
In [24]: df1.duplicated().sum()
```

```
Out[24]: np.int64(541)
```

```
In [25]: # Drop duplicates  
df1.drop_duplicates(inplace= True)
```

```
In [26]: df1.duplicated().sum()
```

```
Out[26]: np.int64(0)
```

Removing NaN values

```
In [27]: df1.isna().sum()
```

```
Out[27]: name                250  
host id                    0  
host_identity_verified    289  
host name                 404  
neighbourhood group       29  
neighbourhood             16  
lat                       8  
long                     8  
country                  532  
country code             131  
instant_bookable         105  
cancellation_policy       76  
room type                 0  
construction year        214  
price                    247  
service fee              273  
minimum nights           400  
number of reviews        183  
last review              15832  
dtype: int64
```

```
In [28]: df1.drop(columns= ['last review'] , inplace= True)
```

```
In [29]: df1.isna().sum()
```

```
Out[29]: name                250
         host id              0
         host_identity_verified 289
         host name            404
         neighbourhood group   29
         neighbourhood         16
         lat                   8
         long                   8
         country               532
         country code          131
         instant_bookable      105
         cancellation_policy    76
         room type              0
         construction year      214
         price                  247
         service fee            273
         minimum nights         400
         number of reviews      183
         dtype: int64
```

Dropping NaN

```
In [30]: df1.dropna(inplace= True)
```

```
In [31]: df1.isna().sum()
```

```
Out[31]: name                0
         host id              0
         host_identity_verified 0
         host name            0
         neighbourhood group   0
         neighbourhood         0
         lat                   0
         long                   0
         country               0
         country code          0
         instant_bookable      0
         cancellation_policy    0
         room type              0
         construction year      0
         price                  0
         service fee            0
         minimum nights         0
         number of reviews      0
         dtype: int64
```

Cleanig individual columns

```
In [32]: df1
```

Out[32]:

	name	host id	host_identity_verified	host name	neighbourhood group	neig
0	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	
1	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	
5	Large Cozy 1 BR Apartment In Midtown East	45498551794	verified	Michelle	Manhattan	
7	BlissArtsSpace!	90821839709	unconfirmed	Emma	Brooklyn	
...	...	...	...	...	...	
102029	Cozy room in a 2B with backyard access	5155155913	unconfirmed	Cristina	Queens	
102030	An East Williamsburg Delight	71564706996	verified	Petra	Brooklyn	
102031	Cozy bedroom in williamsburg	83101851929	verified	Russell	Brooklyn	V
102032	Spacious Room w/ 2 Bed	69037598918	unconfirmed	Christine	Queens	
102040	Room in Queens, NY, near LGA.	56457739998	verified	Sonia	Queens	E

99342 rows × 18 columns



```
In [33]: df1["host_identity_verified"] = df1.host_identity_verified.str.capitalize() # Con
```

```
In [34]: df1
```

Out[34]:

	name	host id	host_identity_verified	host name	neighbourhood group	neig
0	Clean & quiet apt home by the park	80014485718	Unconfirmed	Madaline	Brooklyn	
1	Skylit Midtown Castle	52335172823	Verified	Jenna	Manhattan	
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	Verified	Lyndon	Manhattan	
5	Large Cozy 1 BR Apartment In Midtown East	45498551794	Verified	Michelle	Manhattan	
7	BlissArtsSpace!	90821839709	Unconfirmed	Emma	Brooklyn	
...	...	...	...	...	...	...
102029	Cozy room in a 2B with backyard access	5155155913	Unconfirmed	Cristina	Queens	
102030	An East Williamsburg Delight	71564706996	Verified	Petra	Brooklyn	
102031	Cozy bedroom in williamsburg	83101851929	Verified	Russell	Brooklyn	V
102032	Spacious Room w/ 2 Bed	69037598918	Unconfirmed	Christine	Queens	
102040	Room in Queens, NY, near LGA.	56457739998	Verified	Sonia	Queens	E

99342 rows × 18 columns



In [35]: df1.head()

Out[35]:

	name	host id	host_identity_verified	host name	neighbourhood group	neighbour
0	Clean & quiet apt home by the park	80014485718	Unconfirmed	Madaline	Brooklyn	Kensir
1	Skylit Midtown Castle	52335172823	Verified	Jenna	Manhattan	Mic
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	Verified	Lyndon	Manhattan	East H
5	Large Cozy 1 BR Apartment In Midtown East	45498551794	Verified	Michelle	Manhattan	Murra
7	BlissArtsSpace!	90821839709	Unconfirmed	Emma	Brooklyn	Bec Stuyv

```
In [36]: df1["instant_bookable"] # converting False=0 and True=1
```

Out[36]:

0	FALSE
1	FALSE
4	FALSE
5	TRUE
7	FALSE
...	
102029	FALSE
102030	TRUE
102031	TRUE
102032	FALSE
102040	TRUE

Name: instant_bookable, Length: 99342, dtype: object

```
In [37]: df1["instant_bookable"].apply(lambda x : 1 if x == True else 0)
```

Out[37]:

0	0
1	0
4	0
5	0
7	0
..	
102029	0
102030	0
102031	0
102032	0
102040	0

Name: instant_bookable, Length: 99342, dtype: int64

```
In [38]: df1["instant_bookable"] = df1["instant_bookable"].apply(lambda x : 1 if x == True else 0)
```

```
In [39]: df1 # We can observe that index are not in proper sequence
```

Out[39]:

	name	host id	host_identity_verified	host name	neighbourhood group	neig
0	Clean & quiet apt home by the park	80014485718	Unconfirmed	Madaline	Brooklyn	
1	Skylit Midtown Castle	52335172823	Verified	Jenna	Manhattan	
4	Entire Apt: Spacious Studio/Loft by central park	92037596077	Verified	Lyndon	Manhattan	
5	Large Cozy 1 BR Apartment In Midtown East	45498551794	Verified	Michelle	Manhattan	
7	BlissArtsSpace!	90821839709	Unconfirmed	Emma	Brooklyn	
...	...	...	...	...	...	...
102029	Cozy room in a 2B with backyard access	5155155913	Unconfirmed	Cristina	Queens	
102030	An East Williamsburg Delight	71564706996	Verified	Petra	Brooklyn	
102031	Cozy bedroom in williamsburg	83101851929	Verified	Russell	Brooklyn	V
102032	Spacious Room w/ 2 Bed	69037598918	Unconfirmed	Christine	Queens	
102040	Room in Queens, NY, near LGA.	56457739998	Verified	Sonia	Queens	E

99342 rows × 18 columns



```
In [40]: # Reseting index
df1.reset_index(inplace=True)
```

```
In [41]: df1
```

Out[41]:

	index	name	host id	host_identity_verified	host name	neighbourhood
	0	Clean & quiet apt home by the park	80014485718	Unconfirmed	Madaline	Brooklyn
	1	Skylit Midtown Castle	52335172823	Verified	Jenna	Manhattan
	2	Entire Apt: Spacious Studio/Loft by central park	92037596077	Verified	Lyndon	Manhattan
	3	Large Cozy 1 BR Apartment In Midtown East	45498551794	Verified	Michelle	Manhattan
	4	BlissArtsSpace!	90821839709	Unconfirmed	Emma	Brooklyn
	...	...	...	...	...	...
	99337	Cozy room in a 2B with backyard access	5155155913	Unconfirmed	Cristina	Queer
	99338	An East Williamsburg Delight	71564706996	Verified	Petra	Brooklyn
	99339	Cozy bedroom in williamsburg	83101851929	Verified	Russell	Brooklyn
	99340	Spacious Room w/ 2 Bed	69037598918	Unconfirmed	Christine	Queer
	99341	Room in Queens, NY, near LGA.	56457739998	Verified	Sonia	Queer
	99342					

99342 rows × 7 columns

```
In [42]: df1["instant_bookable"].dtype
```

Out[42]: dtype('int64')


```
In [43]: # Removing $ sign from "price" and "service fee". Also converting them into integer
df1["price"] = df1["price"].str.replace("$", "")
```

```
In [44]: df1["price"]
```

```
Out[44]: 0          966
1          142
2          204
3          577
4         1,060
...
99337       822
99338       455
99339     1,078
99340        103
99341        982
Name: price, Length: 99342, dtype: object
```

```
In [45]: # Removing ', '
df1["price"] = df1["price"].str.replace(", ", "")
```

```
In [46]: df1["price"]
```

```
Out[46]: 0          966
1          142
2          204
3          577
4         1060
...
99337       822
99338       455
99339     1078
99340        103
99341        982
Name: price, Length: 99342, dtype: object
```

```
In [47]: # Removing spaces if any
df1["price"] = df1["price"].str.replace(" ", "")
```

```
In [48]: df1["price"]
```

```
Out[48]: 0          966
1          142
2          204
3          577
4         1060
...
99337       822
99338       455
99339     1078
99340        103
99341        982
Name: price, Length: 99342, dtype: object
```

```
In [49]: type(df1["price"])
```

```
Out[49]: pandas.core.series.Series
```

```
In [50]: # Now Converting "price" column into integer  
  
df1["price"].astype(int).dtype
```

```
Out[50]: dtype('int64')
```

```
In [51]: df1["price"] = df1["price"].astype(int)
```

```
In [52]: df1["price"]
```

```
Out[52]: 0          966  
         1          142  
         2          204  
         3          577  
         4         1060  
         ...  
        99337         822  
        99338         455  
        99339        1078  
        99340         103  
        99341         982  
        Name: price, Length: 99342, dtype: int64
```

```
In [53]: df1["price"].head()
```

```
Out[53]: 0          966  
         1          142  
         2          204  
         3          577  
         4         1060  
        Name: price, dtype: int64
```

```
In [54]: type(df1["price"][1])
```

```
Out[54]: numpy.int64
```

```
In [55]: df1.head()
```

Out[55]:

	index	name	host id	host_identity_verified	host name	neighbourhood group	nei
0	0	Clean & quiet apt home by the park	80014485718	Unconfirmed	Madaline	Brooklyn	
1	1	Skylit Midtown Castle	52335172823	Verified	Jenna	Manhattan	
2	4	Entire Apt: Spacious Studio/Loft by central park	92037596077	Verified	Lyndon	Manhattan	
3	5	Large Cozy 1 BR Apartment In Midtown East	45498551794	Verified	Michelle	Manhattan	
4	7	BlissArtsSpace!	90821839709	Unconfirmed	Emma	Brooklyn	

Now cleaning "service fee" as we did for "price" column

In [56]: `df1.head()`

Out[56]:

	index	name	host id	host_identity_verified	host name	neighbourhood group	nei
0	0	Clean & quiet apt home by the park	80014485718	Unconfirmed	Madaline	Brooklyn	
1	1	Skylit Midtown Castle	52335172823	Verified	Jenna	Manhattan	
2	4	Entire Apt: Spacious Studio/Loft by central park	92037596077	Verified	Lyndon	Manhattan	
3	5	Large Cozy 1 BR Apartment In Midtown East	45498551794	Verified	Michelle	Manhattan	
4	7	BlissArtsSpace!	90821839709	Unconfirmed	Emma	Brooklyn	

In [57]: `df1['service fee']`

```
Out[57]: 0          $193
         1          $28
         2          $41
         3         $115
         4         $212
         ...
        99337        $164
        99338         $91
        99339        $216
        99340         $21
        99341        $196
        Name: service fee, Length: 99342, dtype: object
```

```
In [58]: df1['service fee'] = df1['service fee'].str.replace('$' , '').str.replace(' ' , '')
```

```
In [59]: df1['service fee'] = df1['service fee'].astype(int)
```

```
In [60]: df1['service fee'].dtypes
```

```
Out[60]: dtype('int64')
```

```
In [67]: df1.drop(columns= 'index' , inplace= True)
```

```
In [68]: df1
```

Out[68]:

	name	host id	host_identity_verified	host name	neighbourhood group	neigh
0	Clean & quiet apt home by the park	80014485718	Unconfirmed	Madaline	Brooklyn	
1	Skylit Midtown Castle	52335172823	Verified	Jenna	Manhattan	
2	Entire Apt: Spacious Studio/Loft by central park	92037596077	Verified	Lyndon	Manhattan	E
3	Large Cozy 1 BR Apartment In Midtown East	45498551794	Verified	Michelle	Manhattan	I
4	BlissArtsSpace!	90821839709	Unconfirmed	Emma	Brooklyn	
...	...	...	...	...	...	
99337	Cozy room in a 2B with backyard access	5155155913	Unconfirmed	Cristina	Queens	
99338	An East Williamsburg Delight	71564706996	Verified	Petra	Brooklyn	
99339	Cozy bedroom in williamsburg	83101851929	Verified	Russell	Brooklyn	Wi
99340	Spacious Room w/ 2 Bed	69037598918	Unconfirmed	Christine	Queens	
99341	Room in Queens, NY, near LGA.	56457739998	Verified	Sonia	Queens	Eas

99342 rows × 18 columns



Converting cleanead data into csv format

```
In [69]: df1.to_csv("Cleaned Data.csv")

In [70]: pd.read_csv("Cleaned Data.csv")
```

Out[70]:

	Unnamed: 0	name	host id	host_identity_verified	host name	neighbourh g
	0	Clean & quiet apt home by the park	80014485718	Unconfirmed	Madaline	Broo
	1	Skylit Midtown Castle	52335172823	Verified	Jenna	Manh
	2	Entire Apt: Spacious Studio/Loft by central park	92037596077	Verified	Lyndon	Manh
	3	Large Cozy 1 BR Apartment In Midtown East	45498551794	Verified	Michelle	Manh
	4	BlissArtsSpace!	90821839709	Unconfirmed	Emma	Broo
	...	...	...	...	...	...
	99337	Cozy room in a 2B with backyard access	5155155913	Unconfirmed	Cristina	Qu
	99338	An East Williamsburg Delight	71564706996	Verified	Petra	Broo
	99339	Cozy bedroom in williamsburg	83101851929	Verified	Russell	Broo
	99340	Spacious Room w/ 2 Bed	69037598918	Unconfirmed	Christine	Qu
	99341	Room in Queens, NY, near LGA.	56457739998	Verified	Sonia	Qu

99342 rows × 19 columns



In []:

In []: